Fiscal Year:	FY 2009	**Task Last Updated:**	FY 06/18/2009
PI Name:	Ahmad, Ali Ph.D.		
Project Title:	CogGauge: A Cognitive Assessment Tool		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	32765	**Congressional District:**	24
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	02/12/2009	**End Date:**	02/11/2011
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX08CC18P		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The Cognitive Gauge (CogGauge) tool aims to develop a portable gaming application that assesses cognitive state of astronaut crew members with the goal of determining probable causes of observed cognitive deficits. CogGauge, while engaging astronauts in an entertaining experience, combines predictive tools for assessing cognitive workload with metrics that assess performance decrements. CogGauge uses a hybrid approach combining predictive workload values with behavioral/performance-based workload assessment across a number of task difficulty levels. This comprehensive approach takes into consideration learning effects across a number of cognitive tasks (i.e., mini-games in the gaming context) and derives assessment of performance decrements related to cognitive deficits to identify probable causes of cognitive decrement. Feedback from CogGauge may be provided to astronauts and/or flight surgeons to determine impact on space flight and missions. POTENTIAL NASA COMMERCIAL APPLICATIONS: CogGauge, once developed, forms a foundation for several potential tools for use within NASA. For example, the underlying framework that utilizes a combination of predictive		

	workload and performance/behavioral measures can be used to build several tools that could be integrated within shuttle (or its future replacements) controls to assess astronaut cognitive workload during a mission. Another possible expansion is to assess ground operations centers workload while monitoring a mission progress. In essence, the concepts/approach implemented in CogGauge could potentially expand to several applications within NASA that involve intense human-system integration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In terms of other government and commercial markets, CogGauge could be potentially tailored for use by aircraft pilots to assess their cognitive workload before/after long flying hours. In many situations, the approach/algorithms implemented in CogGauge could be used to model an operator interacting with a complex system, such as Command and Control, Power Plant Control Units, Nursing Stations, and so on. In today's net-centric work environment, operators are required to consume multiple streams of data simultaneously from several monitor stations, which can lead to issues of cognitive overload, decreased performance, and loss of situation awareness. CogGauge could be adapted to monitor operator workload in real-time and drive potential adaptations to displays/task demands to minimize times of overload. Such an extension of CogGauge may be useful in domains such as airport security checkpoints, customs, police stations, etc.
Task Progress:	New project for FY2009. Reporting not required for this SBIR Phase 2 project.
Bibliography Type:	Description: (Last Updated:)